

Probability is the language of decisions

1 A probability is a track record

“70% chance of rain” means: of days like this, about 7 in 10 get rain. You can check it.



“70%” → 7 days in 10

2 Always ask: among what?

$P(\text{attack})$ and $P(\text{attack} \mid \text{this score})$ are different numbers. The bar means “given”: among cases like this one.

3 Base rates rule

A 99%-accurate detector for a 1-in-1,000 event still raises mostly false alarms.

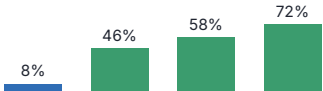
99 real

4,995 false

$$P(\text{attack} \mid \text{alert}) \approx 2\%$$

4 Evidence multiplies the odds

Each clue multiplies the odds by its likelihood ratio. That’s Bayes’ rule, in one line.



5 Precision and recall

Accuracy hides the rare class. “Never alert” scores 92% here.



6 Small samples wobble

3 attacks in 20 alerts could be a 4% rate or a 35% one. Report the range, not just the number.